

Event-triggered voltage regulation and current sharing in neighbors-based second control for DC microgrids

Yi Wei Feng¹  and Yu Hang Li¹

Abstract

This paper explores a distributed cooperative control method based on event-triggering for voltage regulation and current sharing in DC microgrids. The method calculates the average voltage using a voltage observer and employs secondary control to synchronize the voltage of each Distributed Energy Resource (DER) to the desired value, achieving current sharing among the DERs. The proposed scheme relies solely on information exchange between adjacent DERs and uses an event-triggering condition (i.e. measurement error reaching a specified threshold) to determine controller updates, thereby reducing unnecessary communication. The stability of the approach is verified using Lyapunov functions, and the effectiveness of the control strategy is confirmed through Matlab/Simulink. Results show that the method exhibits good stability and robustness under load changes and system structure changes, significantly reducing the number of communications compared to traditional periodic control methods.

Keywords

DC microgrids, event-triggered control, distributed control, voltage regulation, threshold, current sharing

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Introduction

With the increasing demand for renewable energy sources (such as electricity), microgrids are gradually developing as a form of distributed renewable energy sources. Compared to traditional AC microgrids, DC microgrids do not store reactive currents and do not need to address problems related to frequency and power quality. Their high efficiency, reliability, and simple control structure make them more attractive.¹

The two main control objectives of a DC microgrid are voltage regulation and current sharing.² Voltage regulation involves adjusting the bus voltages of all DERs in according to the voltage reference point specified by the tertiary control. Current sharing means achieving a balanced distribution of currents among all DERs and enabling current exchange to avoid circulating currents.³ To achieve these two control objectives, numerous research works have been reported, with the hierarchical control structure being the most well-known.^{4,5} Specifically, the hierarchical control of a DC microgrid consists of three levels: primary, secondary, and tertiary control. The primary control uses droop control to achieve voltage regulation and load sharing. Due to the existence of DC bus voltage deviation between multiple DERs, the secondary control strategy is employed to correct the bus voltage deviation.⁶

Additionally, the tertiary control is responsible for adjusting power allocation to achieve economic dispatch of the entire system.⁷ This paper focuses on the secondary control strategy for microgrids. It should be noted that centralized secondary control was previously used to achieve voltage stabilization.⁸ In microgrids, a centralized controller is used to monitor and control the voltage and current values of each DER. However, centralized secondary control requires a complex communication network and has poor scalability, and making it unsuitable for large-scale microgrid systems.⁹ Distributed secondary control overcomes these drawbacks by exchanging information only with neighboring units, reducing energy waste, and making the system more efficient, reliable, and structurally simpler. It is widely used in DC microgrid systems with limited bandwidth.¹⁰ A distributed secondary voltage control method has been proposed for DC microgrids, which

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achieves voltage regulation and current sharing by exchanging information about the DC bus voltage and its neighboring DER.¹¹ In,¹² a cooperative control method based on multiagent system consensus was studied to achieve voltage stabilization and precise current allocation. In,¹³ a distributed secondary control scheme based on a multiagent system was designed to achieve voltage restoration and current balance accuracy. In,¹⁴ the consensus theory of multi-agent systems was introduced and a distributed cooperative control scheme based on multi-agent systems was proposed to realize energy management and economic dispatch of all DERs through a sparse communication network in DC microgrids. In these secondary control methods, each DER must constantly communicate with neighboring units. It is a well-known fact that excessive communication leads to energy waste and competition for communication resources among agents. Particularly in large-scale DC microgrids with multiple control objectives, communication networks are prone to traffic congestion, communication delays, and cyber attack,¹⁵ which can affect system stability.

Event-triggered control (ETC) is a strategy aimed at reducing communication overhead. It only activates control actions and triggers necessary data exchanges when predefined conditions are met, thereby avoiding unnecessary communication and computational resource waste. To alleviate the communication burden and prevent communication redundancy and transmission congestion in microgrids, event-triggered communication mechanisms are employed in microgrids to reduce the amount of data exchanged in the communication network. This has become a popular topic and has been extensively studied.¹⁶ In,¹⁷ an ETC method was proposed and an auxiliary centralized controller is designed to achieve voltage control; however, the centralized controller requires a more complex communication network to monitor the state information of each unit, which in turn increases the communication burden once again. By introducing an event-triggered distributed controller based on the ac microgrid,¹⁸ it was applied to achieve precise power distribution in AC microgrids. In,¹⁹ a distributed secondary frequency control strategy based on dynamic event-triggering was proposed. This strategy applies a distributed ETC mechanism to AC microgrid systems. The controller updates the output of the DG units when the sampling error of voltage or frequency exceeds a certain threshold. The work focused to AC microgrids, while in DC microgrids,²⁰ a distributed control strategy for DC microgrids, combining event-triggered average consensus protocols and fractional-order proportional-integral local controllers to achieve voltage stabilization and energy balancing of energy storage systems. This reduces the amount of data exchanged between the converters. However, the previous control scheme did not consider Zeno Behavior. Zeno Behavior refers to

the phenomenon where event triggering mechanism occurs an finite period, potentially causing system instability. Literature²¹ Proposed a a mixed time-state dependent Event-triggering strategy, the strategy was developed to achieve global voltage regulation and proportional distribution of output current, while also eliminating Zeno behavior.

Compared to the existing work, the contributions of this paper are as follows:

1. The control scheme proposed in this paper effectively achieves global voltage regulation and current sharing. Instead of requiring a fully connected communication network among all DERs, the scheme only necessitates exchanging and computing state information with neighboring units, thereby ensuring the reliability of the control strategy.
2. This paper designs an undirected graph consensus control protocol with a virtual leader for DC microgrid systems and calculates the conditions for distributed event triggering. Experimental results show that, compared with periodic strategies, the control strategy involves fewer controller updates and effectively reduces the communication burden on the controllers.
3. The stability of the control strategy was proven using the Lyapunov function, and the lower bound of the time interval between events was estimated to prevent Zeno behavior.

The rest of this paper is organized as follows. First, we introduce the basic concepts of graph theory. This is followed by a detailed description of the system model and control objectives for DC microgrids. We then design a distributed ETC protocol and discuss its stability. The effectiveness of the proposed control strategy is verified through simulation. Finally, we summarize the contributions of this paper and outline future research directions.

System model

Graph theory

The communication network topology between each DER of the DC microgrid shown in Figure 1 can be represented by an undirected graph $G = (v, \varepsilon)$, where $v = (v_1, v_2, \dots, v_N)$ denotes the finite set of nodes of a communication network where any edge consists of two distinct nodes. Edge set $\varepsilon \subseteq v \times v$, edge $(v_i, v_j) \in \varepsilon$ indicates that a node can receive information from node v_i , where nodes v_i and v_j are neighbors of each other, and the undirected graph can be introduced $(v_j, v_i) \in \varepsilon$, so the neighbor set of node v_i can be expressed as $N_i = \{j \in v | (v_i, v_j) \in \varepsilon\}$. The adjacency matrix $A = (a_{ij})_{N \times N}$ represents the communication network between agents, a_{ij} defined as

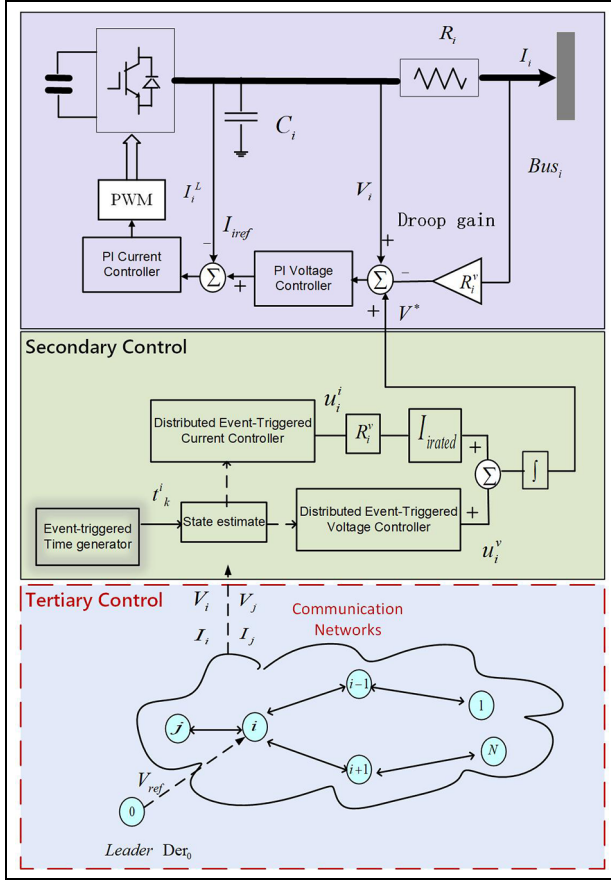


Figure 1. Block diagram of distributed cooperative control based on event triggering.

$$a_{ij} = \begin{cases} 1, & \text{if } (v_i, v_j) \in \varepsilon \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

The Laplace matrix of an undirected graph is defined as $L = D - A$, $L = (l_{ij})_{N \times N}$, $l_{ij} = \begin{cases} -a_{ij}, & i \neq j \\ \sum_{i=1}^N a_{ij}, & i = j \end{cases}$. The degree matrix $D = \text{diag}(d_1, d_2, \dots, d_N)$ is the diagonal matrix. The leader unit Der_0 , which provides the voltage reference value, represents the transmission path to other units. This is represented with an adjacency matrix $F_i = \text{diag}(f_1, f_2, \dots, f_N)$, $f_i = 1$ indicates that the reference voltage value can be received from Der_0 ; otherwise, $f_i = 0$.

DC microgrid model

This paper introduces the DC microgrid system model, the overall structure of which is depicted in Figure 2. It shows N converter interfaces connecting DERs in parallel to a common coupling point via connecting wires. A hierarchical control scheme is designed for the islanded DC microgrid, and basic concepts of graph theory are introduced.

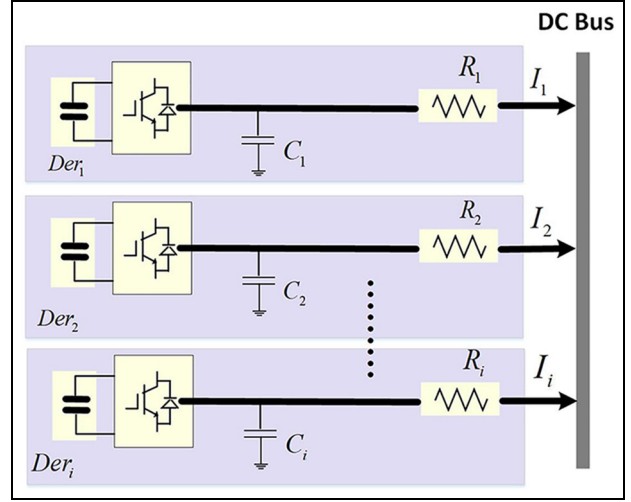


Figure 2. DC microgrid system structure.

In the physical layer as shown in Figure 1, voltage control and current regulation are realized by using droop-based control as primary control. The local voltage reference V_i be expressed as

$$V_i = V^* - R_i^v I_{i, \text{rated}} I_i \quad (2)$$

Where V^* denotes the voltage set point, R_i^v denotes the virtual resistance of Der_i , and I_i is the out current from Der_i .

Our two main objectives in designing the controller are voltage regulation and current sharing. voltage regulation can be expressed as

$$\lim_{t \rightarrow +\infty} |\bar{V}_i(t) - V_{ref}| = 0 \quad (3)$$

$$\lim_{t \rightarrow +\infty} |\bar{V}_j(t) - \bar{V}_i(t)| = 0 \quad (4)$$

The average voltage $\bar{V}_i$ of each DER unit can reach the desired value V_{ref} while ensuring current sharing.

$$\lim_{t \rightarrow +\infty} |I_j(t) - I_i(t)| = 0 \quad (5)$$

To compensate for the voltage deviation caused by the droop control, a secondary correction factor needs to be introduced to adjust the voltage and current. From equation (4), the secondary control input is given by

$$\dot{V}_i(t) = \dot{V}^* - R_i^v I_{i, \text{rated}} \dot{I}_i = u_i^v(t) \quad (6)$$

$$\dot{I}_i(t) = u_i^i(t) \quad (7)$$

The average voltage regulation requires the introduction of a voltage observer to estimate the average voltage of individual unit, The estimated average voltage of DER is

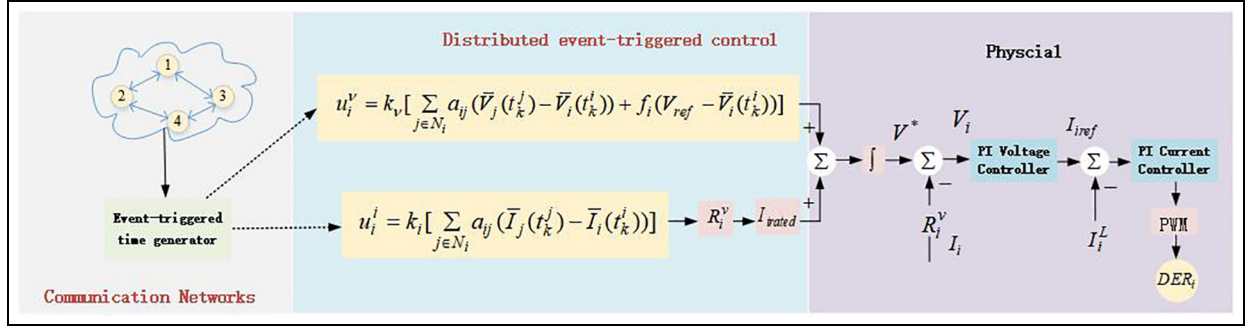


Figure 3. Distributed event triggering control structure.

$$\bar{V}_i(t) = V_i(t) + \int_0^t \sum_{j \in N_i} a_{ij} (\bar{V}_j(t) - \bar{V}_i(t)) dt \quad (8)$$

Event-triggered distributed secondary control

Distributed controller design

By considering Der_i to exchange information with neighboring units, we can derive the distributed voltage consistency controller as

$$\bar{V}_i(t) = u_i^v(t) = k_v \left[\sum_{j \in N_i} a_{ij} (\bar{V}_j(t) - \bar{V}_i(t)) + f_i (V_{ref} - \bar{V}_i(t)) \right] \quad (9)$$

k_v is the equalized current control gain, a_{ij} denotes the communication weight, f_i denotes whether each DER can receive the voltage reference value V_{ref} from the virtual leader, $F = \text{diag}\{f_i\}$, and it is requires that at least one node can receive the reference voltage value from Der_0 .

Similarly, the distributed current consistency controller of DER is

$$u_i^i(t) = k_i \left[\sum_{j \in N_i} a_{ij} (\bar{I}_j(t) - \bar{I}_i(t)) \right] \quad (10)$$

k_i is the equalized current control gain. Finally, using the voltage control input u_i^v and the current control input u_i^i as described above, the voltage setpoint V^* based on the droop primary control (3) can be calculated as

$$V^* = \int (u_i^v + R_i^v I_{rated} u_i^i) \quad (11)$$

Distributed control with event-triggered communication

Due to the introduction of the event-triggered mechanism in the secondary control inputs, the detailed

structure of the event-triggered controller is shown in Figure 3. Each DER can only sample and transmit information in a discrete trigger time sequence. The expression of the discrete trigger time sequence is as follows

$$t_0^i = 0 < t_1^i < t_2^i < \dots < t_k^i, k \in N \quad (12)$$

t_k^i is not a periodic sequence of sampling times, but rather the k_{th} triggering time at which Der_i satisfies the triggering condition. For each DER unit, only at each $t = t_k^i$ measure their own voltage and current value $V_i(t_k^i)$, $I_i(t_k^i)$ of their neighbors, and them unchanged in $t \in [t_k^i, t_{k+1}^i)$. The trigger is activated again at the time.

As shown in Figure 3, an event-triggered voltage controller based on event triggering can be designed as

$$u_i^v(t_k^i, t_k^i) = k_v \left[\sum_{j \in N_i} a_{ij} (\bar{V}_j(t_k^j) - \bar{V}_i(t_k^i)) + f_i (V_{ref} - \bar{V}_i(t_k^i)) \right] \quad (13)$$

$t \in [t_k^i, t_{k+1}^i)$, t_k^i denotes the k_{th} trigger time.

Similarly, the event-triggered current-based controller is designed as

$$u_i^i(t_k^i, t_k^i) = k_i \left[\sum_{j \in N_i} a_{ij} (\bar{I}_j(t_k^j) - \bar{I}_i(t_k^i)) \right] \quad (14)$$

Utilizing the designed ETC strategy, the distributed cooperative control of each DER unit is achieved, as illustrated in Figure 3, which provides the detailed design scheme of the main controller and the distributed secondary controller.

For ease of computation, define $\bar{x}_i(t)$, $\dot{x}_i(t)$ represent $\bar{V}_i(t)$, $\dot{V}_i(t)$. The dynamics of each DER can be described as follows

$$\dot{\bar{x}}_i(t) = u_i(t_k^i, t_k^i) = k_v \left[\sum_{j \in N_i} a_{ij} (\bar{x}_j(t_k^j) - \bar{x}_i(t_k^i)) - f_i \bar{x}_i(t_k^i) \right] \quad (15)$$

$x_i \in R$ indicates the state, and $u_i \in R$ is the control input.

Using the Laplace matrix, the closed-loop system of (15) is represented as

$$\dot{\bar{x}}_i(t) = u_i(t_k^j, t_k^i) = -k_v(L + F)\bar{x}_i(t_k^i, t_k^j) \quad (16)$$

Define the state measurement error of Der_i as

$$e_i(t) = \bar{x}_i(t_k^i, t_k^j) - \bar{x}_i(t), t \in (t_k^i, t_{k+1}^i) \quad (17)$$

Theorem 1. Consider a secondary controller consisting of $\dot{x} = u_i$. If the undirected communication topology graph G is connected, then the distributed controller (16) can keep all follower agents following the leader node with asymptotic consistency under the control condition (19).

To achieve the overall goal of event triggering, we design and analyze an ETC scheme to address the consistency problem in (16).

The trigger condition for error e is designed as

$$t_{k+1}^i = \inf\{t > t_k^i : \|e_i(t)\| > \xi_i \|h_i(t)\|\} \quad (18)$$

$\|\cdot\|$ denotes the Euclidean norm of the vector, $\inf\{\cdot\}$ denotes the lower definite function, where $0 < \xi_i < \frac{-2a}{2a|N_i| + 2af_i - a^2 - 1}$, $0 < a < 1$, $h_i(t) = (L + F)\bar{x}_i(t)$.

Proof: Considering the ISS Lyapunov candidate function $W = \frac{1}{2}x^T Lx$, combined with the measurement error (17), we have

$$\begin{aligned} \dot{W}(\bar{x}) &= \bar{x}^T(L + F)\dot{\bar{x}} \\ &= -k_v \bar{x}_i(t)^T(L + F)^2(\bar{x}_i(t) + e_i(t)) \end{aligned} \quad (19)$$

Which can be expressed as

$$\dot{W} = -k_v h_i(t)^T h_i(t) - k_v h_i(t)^T(L + F)e_i(t) \quad (20)$$

The error of the Laplace matrix can be written as

$$\dot{W} = -k_v \sum_{i=1}^N h_i^2 - k_v \sum_{I=1}^n \sum_{j \in N_i} h_i(e_i - e_j) - k_v \sum_{i=1}^N h_i f_i e_i \quad (21)$$

Since the communication network undirected graph is symmetric, then

$$\sum_{i=1}^N \sum_{j \in N_i} X_i Y_i = \sum_{i=1}^N |N_i| X_i Y_i \quad (22)$$

Therefore (21) can be written as

$$\begin{aligned} \dot{W} &= -k_v \sum_{i=1}^N h_i^2 - k_v \sum_{i=1}^N |N_i| h_i e_i \\ &\quad + k_v \sum_{i=1}^N |N_i| h_i e_j - k_v \sum_{i=1}^N h_i f_i e_i \end{aligned} \quad (23)$$

By the nature of the paradigm, it can be expressed as

$$\begin{aligned} \dot{W} &\leq -k_v \sum_{i=1}^N h_i^2 - k_v \sum_{i=1}^N |N_i| \|h_i\| \|e_i\| \\ &\quad + k_v \sum_{i=1}^N \sum_{j \in N_i} \|h_i\| \|e_j\| - k_v \sum_{i=1}^N f_i \|h_i\| \|e_i\| \end{aligned} \quad (24)$$

Combining equations (18) and (24), by calculating

$$\begin{aligned} \dot{W} &\leq -k_v \sum_{i=1}^N (1 + \xi_i |N_i|) \|h_i\|^2 \\ &\quad + k_v \sum_{i=1}^N \sum_{j \in N_i} \xi_j \|h_i\| \|h_j\| - k_v \sum_{i=1}^N \xi_i f_i \|h_i\|^2 \end{aligned} \quad (25)$$

Using the inequality $|xy| \leq (\frac{a}{2})x^2 + (\frac{1}{2a})y^2$, $a > 0$, (25) can be written as

$$\begin{aligned} \dot{W} &\leq -k_v \sum_i (1 + \xi_i |N_i| + f_i \xi_i) \|h_i\|^2 + k_v \\ &\quad \sum_{i=1}^N \sum_{j \in N_i} \frac{a}{2} \xi_j \|h_i\|^2 + k_v \sum_{i=1}^N \sum_{j \in N_i} \frac{1}{2a} \xi_j \|h_i\|^2 \end{aligned} \quad (26)$$

According to the symmetry property of the undirected graph G and (22), we can get

$$\sum_{i=1}^N \sum_{j \in N_i} \frac{1}{2a} \|h_j\|^2 = \sum_{i=1}^N \frac{1}{2a} |N_i| \|h_i\|^2 \quad (27)$$

Using (27), the form of (26) can be written as

$$\begin{aligned} \dot{W} &\leq -k_v \\ &\quad \frac{\sum_{i=1}^N [2a(1 + \xi_i |N_i| + f_i \xi_i) \|h_i\|^2 + \sum_{i=1}^N (\xi_i a^2 + \xi_i) \|h_i\|^2]}{2a} \end{aligned} \quad (28)$$

Given the trigger condition is

$$0 < \xi_i < \frac{-2a}{2a|N_i| + 2af_i - a^2 - 1} \quad (29)$$

Then

$$\dot{W} \leq 0 \quad (30)$$

This demonstrates that the execution condition triggered by the event (19) ensures asymptotic stabilization of $x(t)$ for all agents, that is, $\lim_{x \rightarrow \infty} \|\bar{x}_i(t)\| = \lim_{x \rightarrow \infty} \|\bar{V}_i(t) - V_{ref}\| = 0$. The output voltages of all DERs are finally synchronized to the reference voltage, and the event-triggered voltage controller (14) can achieve voltage regulation of the DC microgrid system.

The stability analysis of the current sharing event-triggered controller is similar to the above derivation process. The u_i^i in equation (15) differs from u_i^v in equation (14) in that it does not have the pinning gain f_i of the virtual leader, thus requiring only a simple modification to prove its stability as well. We will then analyze the Zeno-free behavior of the controller.

Theorem 2. Consider the system $\bar{V}_i = u_i^v$ under the control rate (13). The communication network is an undirected topology graph, assuming that one node has access to the leader node, and under the trigger condition (18), all units do not exhibit Zeno behavior, and the event interval τ is strictly positive.

From (17), the derivative of the state measurement error is

$$\dot{e}_i = -\dot{\bar{x}}_i = k_v(L + F)\bar{x}_i(t_k^i, t_k^j) \quad (31)$$

Proof. Consider the following time derivatives

$$\begin{aligned} \frac{d\|e\|}{dt\|\bar{x}\|} &= \frac{d}{dt} \left(\frac{\|e\|}{\|(L + F)\bar{x}\|} \right) \\ &= \frac{\|\bar{x}\| \|e\|^{-1} e^T \dot{e} - \|e\| \|\bar{x}\|^{-1} \bar{x}^T \dot{\bar{x}}}{\|\bar{x}\|^2} \\ &= -\frac{\|e\|^{-1} e^T \dot{\bar{x}}}{\|\bar{x}\|} - \frac{\|e\| \bar{x}^T \dot{\bar{x}}}{\|\bar{x}\|^3} \\ &\leq \frac{\|e\| \|\dot{\bar{x}}\|}{\|\bar{x}\| \|e\|} + \frac{\|e\| \|\bar{x}\| \|\dot{\bar{x}}\|}{\|\bar{x}\|^3} \\ &\leq \frac{\|\dot{\bar{x}}\|}{\|\bar{x}\|} \left(1 + \frac{\|e\|}{\|\bar{x}\|} \right) \end{aligned} \quad (32)$$

Assuming that

$$M = \frac{\|e\|}{\|\bar{x}\|} = \frac{\|e\|}{\|(L + F)\bar{x}\|} \quad (33)$$

Then

$$\dot{M} \leq \frac{\|k_v(L + F)\bar{x}_i(t_k^i, t_k^j)\|}{\|\bar{x}\|} (1 + M) \quad (34)$$

From (18) we have $\bar{x}_i(t_k^i, t_k^j) = e + \bar{x}$, and we can get

$$\begin{aligned} \dot{M} &\leq \frac{k_v \|L + F\| \|e + \bar{x}\|}{\|\bar{x}\|} (1 + M) \\ &\leq k_v \|L + F\| \frac{\|e + \bar{x}\|}{\|\bar{x}\|} (1 + M) \\ &\leq k_v \|L + F\| \frac{\|e\| + \|\bar{x}\|}{\|\bar{x}\|} (1 + M) \\ &\leq k_v \|L + F\| \left(1 + \frac{\|e\|}{\|\bar{x}\|} \right) (1 + M) \\ &\leq k_v \|L + F\| (1 + M)^2 \end{aligned} \quad (35)$$

Which $\varphi = k_v \|L + F\|$

$$\dot{M} \leq \varphi (1 + M)^2 \quad (36)$$

Integrating both sides of (35), we get

$$\int_0^{M(\tau, 0)} \frac{dM}{\varphi(1 + M)^2} \leq \int_0^\tau dt \quad (37)$$

First solve for both sides by integrating

$$\int_0^{M(\tau, 0)} \frac{dM}{\xi(1 + M)^2} = \frac{1}{\xi} - \frac{1}{\xi(1 + M(\tau, 0))} \leq \tau \quad (38)$$

The results of the calculations are

$$\tau \geq \frac{M(\tau, 0)}{\varphi(1 + M(\tau, 0))} \quad (39)$$

$M(\tau, 0)$ and φ are both positive, and the interval τ between events is strictly positive, thus avoiding the Zeno behaviour, the proof process is complete.

Simulation results

To verify the effectiveness and reliability of the investigated control strategy, a constructed islanded DC microgrid with five DERs connected in parallel is simulated using MATLAB/Simulink. In this paper, an undirected graph with a virtual leader is considered, and the communication topology of the microgrid is shown in Figure 4. The parameters of each DER and controller are shown in Table 1, which refers to papers^{22,23} when selecting the parameters in the simulation example. The reference voltage value for Der_0 is 48 V.

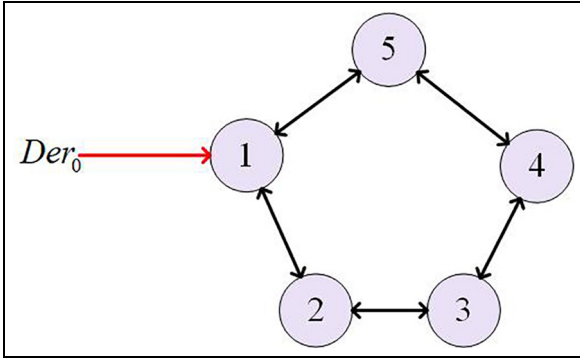
Load change

In this section, we verify the stability of the investigated event-triggered controller under sudden load changes.

The experimental procedure is as follows: at $t=0$ s, the designed secondary controller starts, and at $t=2$ s, the load on Der_3 is doubled. From Figures 5 and 6, all DER bus voltages decrease and the current increases, with the voltage returning to the reference value after 0.5s. At $t=3$ s, the added loads are disconnected from

Table 1. Parameters for the DC microgrid.

Parameters	Der_1	Der_2	Der_3	Der_4	Der_5
V_{DC}	100V	80V	100V	80V	100V
f_{sw}	20kHz	20kHz	20kHz	20kHz	20kHz
L_f	2mH	1.95mH	2mH	1.95mH	2mH
C_f	470uF	420uF	470uF	420uF	470uF
R^v	0.5 Ω	0.2 Ω	0.5 Ω	0.2 Ω	0.5 Ω
R_{load}	21 Ω	25 Ω	23 Ω	27 Ω	
ξ_i	0.0405	0.0426	0.0413	0.0376	0.0395
k_v	2	2	2	2	2
k_i	5	5	5	5	5
	$Line_2$				
$Line_1$	0.1 Ω		$Line_3$	$Line_4$	
0.05 Ω	1.5uH		0.05 Ω	0.1 Ω	
3uH			3uH	1.5uH	

**Figure 4.** Undirected communication graph G.

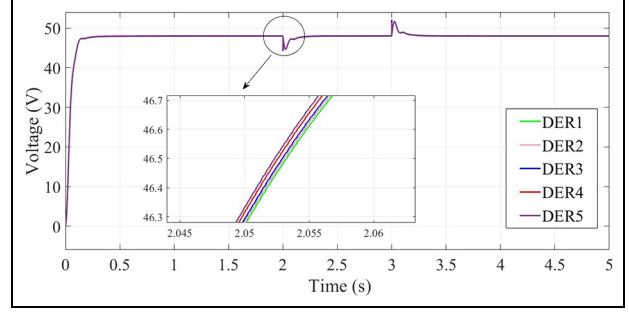
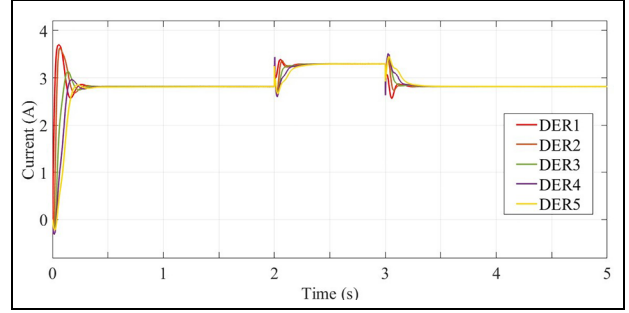
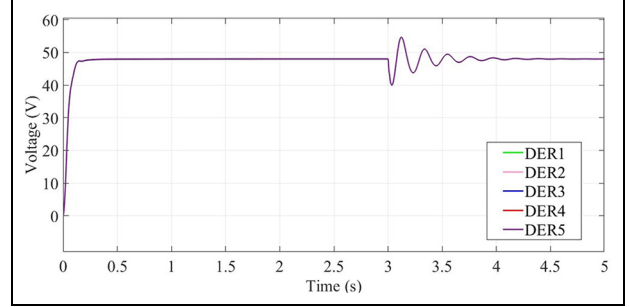
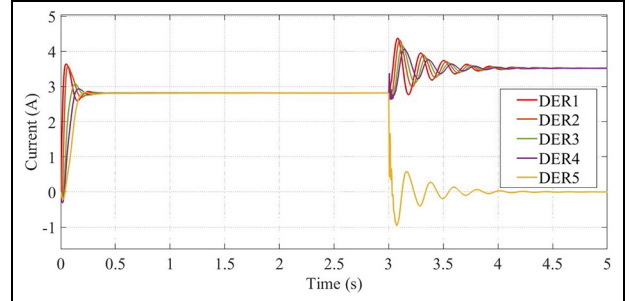
the microgrid, resulting in small fluctuations in all DER bus voltages, and the currents return to the initial values. The results demonstrate that the proposed control scheme is robust; the bus voltage remains unchanged when the load changes, and the output current is reasonably distributed among each DER.

Power system structure change

To verify the robustness of the controller, in this section, we change the structure of the DC microgrid system to test its stability. At $t = 3$ s, Der_5 is disconnected from the grid, as shown in Figures 7 and 8. It can be observed that when Der_5 is disconnected from the DC microgrid, all the DER voltages and currents experience a small fluctuation, but still recovered to the reference value, and all the DERs can still share currents quickly. The results indicate that the controller maintains good stability and effectiveness even when the structure of the power system changes abruptly.

Event-triggered global error

Figures 9 and 10 show the variation values of error $\|e_i(t)\|$ and their thresholds for global voltage and current under the distributed event-triggered controller, with the event-triggered error values for current being

**Figure 5.** Output voltage.**Figure 6.** Output current when the load changes.**Figure 7.** Output voltage when the Power system structure changes.**Figure 8.** Output current when the Power system structure changes.

smaller. The experimental procedure is the same as in Section 4.1, and the error values are close to zero due to the system's nonlinearity. Additionally, because the

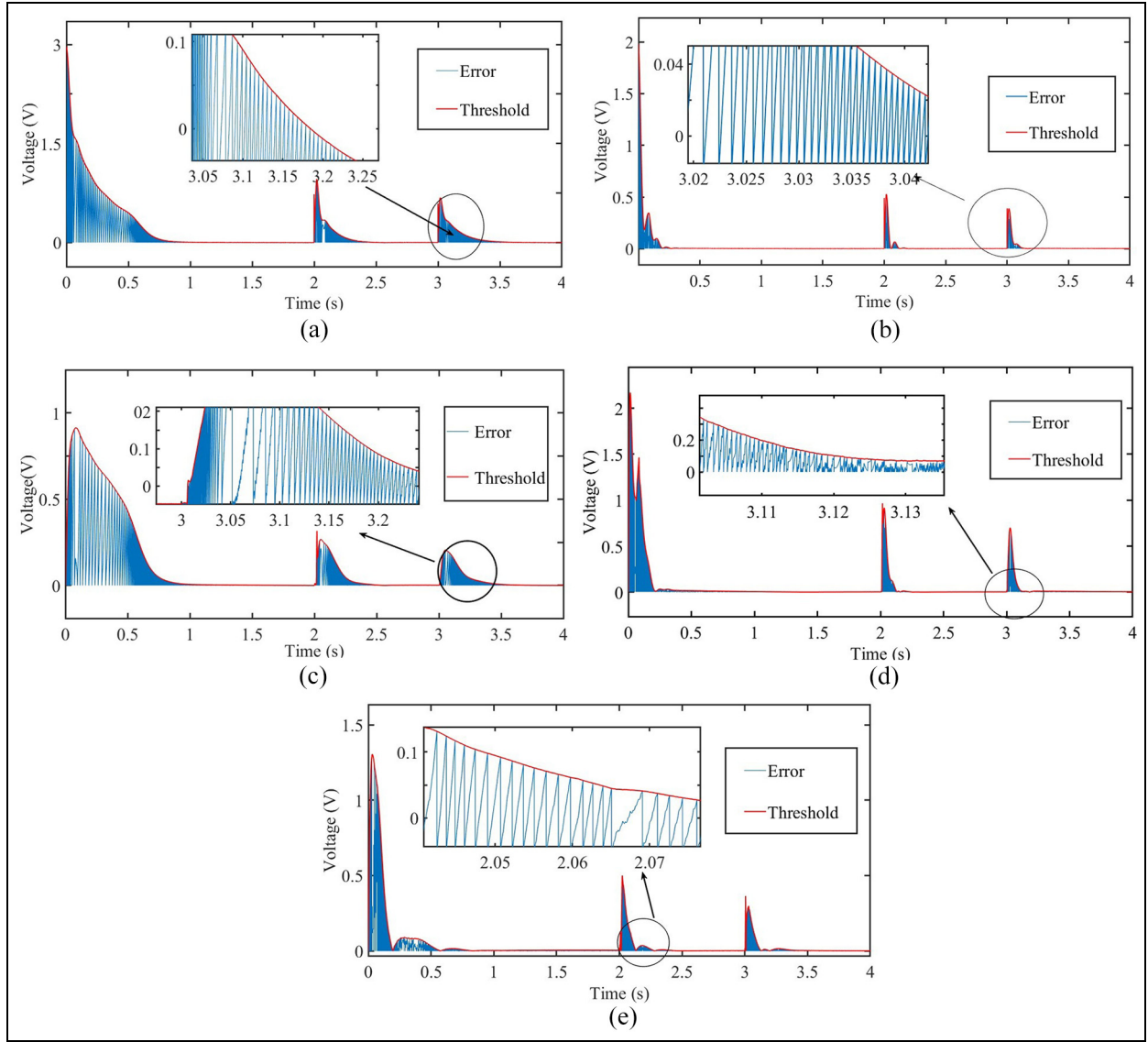


Figure 9. Voltage error with event triggered control.

secondary control scheme is cooperative, the error values of each DER unit are obviously different, and all DERs are triggered individually for sampling communication irregularly and asynchronously, and after reaching stabilization, the error values are almost none and the controller stops updating. Since Der_1 directly receives the reference voltage value from the virtual leader, the number of triggers is higher when starting the secondary controller, and the stabilization is faster, so the number of triggers for each DER is different. As can be seen from Figures 9 and 10, the error value of each DER controller converges to zero asymptotically when the secondary controller is started and the load changes, and then the number of departures becomes more until it stabilizes again.

Comparison with the existing methods

In this section, the ETC strategy designed in this paper is compared with the strategies proposed in the literature,^{20,21} as well as with the periodic communication control strategy adopted in¹⁵. The comparison is based on the number of communication experiments conducted under the same experimental setup parameters but with different control methods. The results, as shown in Figure 11, indicate that the ETC strategy can significantly reduce the communication burden between each DER compared to the periodic control strategy. Furthermore, compared with other ETC methods, the scheme designed in this paper exchanges the least amount of data and more significantly reduces the communication burden.

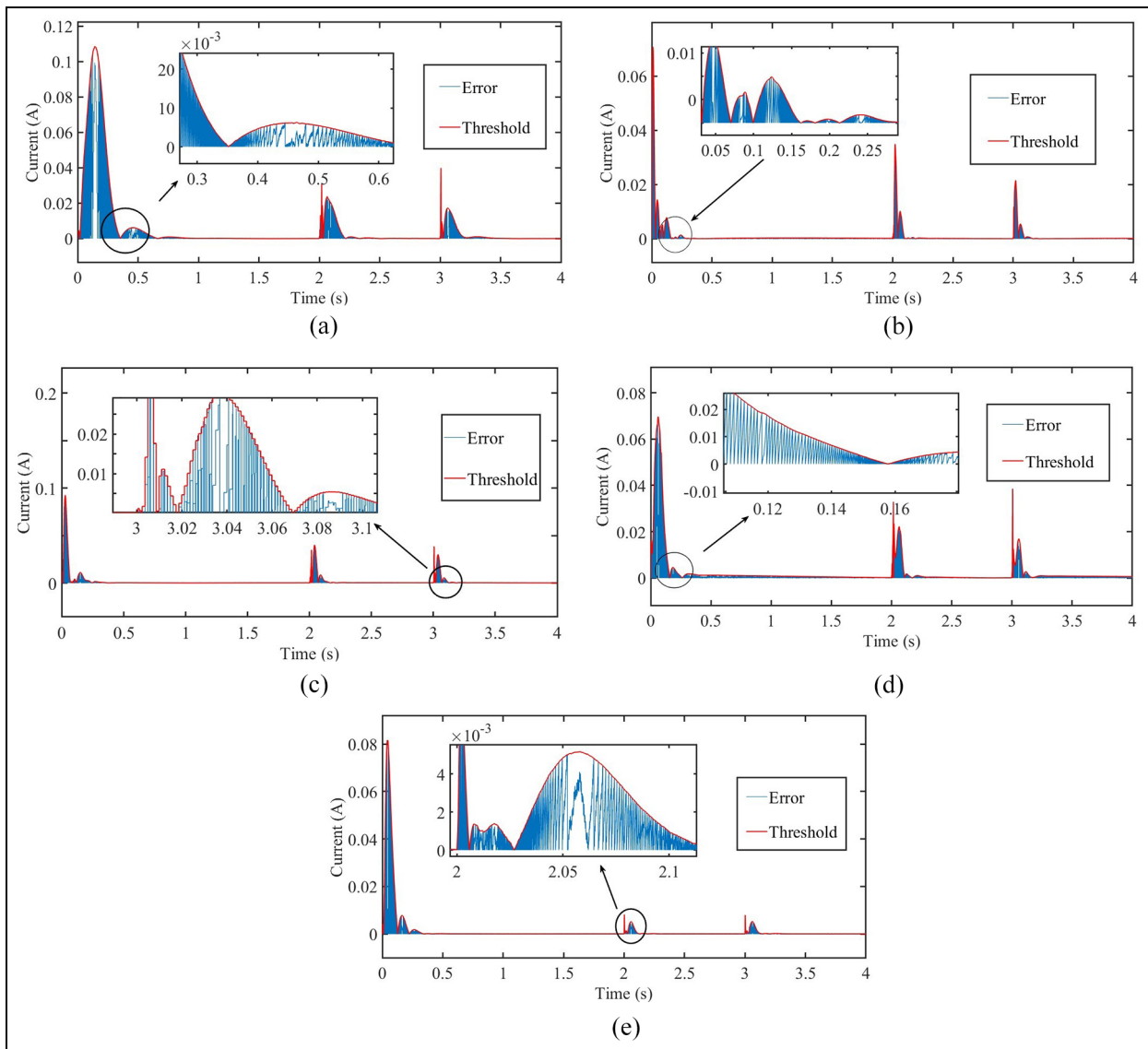


Figure 10. Current error with event triggered control.

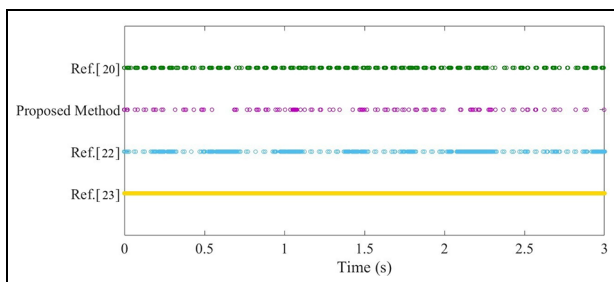


Figure 11. Event-triggered time instants between different methods.

Conclusions

This paper proposes an event-triggered control strategy for microgrids to ensure precise voltage regulation and current sharing through fully distributed control. The event-triggered mechanism reduces communication

burden and its stability and convergence are verified by Lyapunov theory. The derived event-triggering condition ensures the achievement of control objectives without Zeno behavior. The performance of the scheme is validated through various simulation scenarios, demonstrating superior performance and reduced communication burden compared to traditional methods. However, the event triggering is static. Future work will explore dynamic event-triggering and communication delays to enhance robustness against network failures and disturbances.

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Ethical consideration

This article does not contain any studies with human or animal participants.

Informed consent

There are no human participants in this article and informed consent is not required.

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Declaration of conflicting interests

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Data availability statement

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

Trial registration number/date

There are no trial in this article

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